

EDMONTON BLATCHFORD, AB, Canada

WMO: 711570

Lat: 53.5799N

Lon: 113.5156W

Elev: 2201

StdP: 13.56

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -19.1	(c) -13.7	(d) -27.3	(e) 1.3	(f) -18.3	(g) -21.5	(h) 1.8	(i) -13.0	(j) 23.4	(k) 24.1	(l) 20.8	(m) 22.1	(n) 5.9	(o) 210	(p) 0.648

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	83.2	63.9	80.0	62.4	77.1	61.1	66.9	79.1	64.8	76.2	63.0	73.7	9.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.6	92.3	72.3	60.4	85.1	70.1	58.5	79.4	68.2	32.7	79.4	31.1	76.2	29.6	73.7	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	19.1	16.7	-24.1	89.4	6.2	3.4	-28.6	91.8	-32.2	93.8	-35.7	95.7	-40.3	98.2
			-24.4	70.3	6.2	2.4	-28.8	72.1	-32.5	73.5	-35.9	74.9	-40.4	76.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.7	13.7	17.3	26.3	40.7	52.8	60.1	64.7	62.5	53.7	41.4	25.9	15.7
	DBStd	21.20	15.62	13.51	13.47	9.70	8.42	5.87	5.61	6.01	7.89	8.50	12.03	13.83
	HDD50	5267	1124	915	734	300	68	3	0	2	50	286	722	1063
	HDD65	9376	1589	1335	1199	729	385	168	75	118	345	733	1172	1528
	CDD50	1505	0	0	0	21	153	306	456	390	161	18	0	0
	CDD65	139	0	0	0	0	6	20	66	42	5	0	0	0
	CDH74	1405	0	0	0	3	108	201	578	404	107	4	0	0
	CDH80	282	0	0	0	0	16	33	133	81	19	0	0	0
Wind	WSAvg	7.7	6.7	6.8	7.7	8.9	9.0	8.4	8.0	7.4	7.7	7.8	6.9	6.7
Precipitation	PrecAvg	18.9	1.0	0.7	0.7	0.9	1.9	3.2	3.7	2.7	1.7	0.7	0.7	0.9
	PrecMax	24.6	2.9	1.8	2.0	2.1	5.6	7.7	7.3	6.7	4.5	2.6	2.0	2.1
	PrecMin	13.3	0.1	0.1	0.2	0.0	0.2	0.5	1.2	0.3	0.1	0.0	0.0	0.1
	PrecStd	3.0	0.7	0.4	0.5	0.5	1.1	1.8	1.7	1.7	1.0	0.6	0.5	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.2	47.0	56.4	72.8	82.4	84.0	88.3	86.5	82.5	71.5	55.4	47.0
		MCWB	37.5	38.2	43.8	52.4	57.7	62.6	66.8	65.6	61.9	54.7	44.1	38.2
	2%	DB	41.4	41.7	50.2	66.4	76.8	79.2	83.5	81.9	76.7	63.1	47.9	41.4
		MCWB	35.6	34.9	39.7	48.8	55.7	59.9	65.7	64.3	59.3	50.0	40.1	35.2
	5%	DB	38.2	38.3	46.3	61.0	72.6	75.7	80.1	78.4	72.3	57.9	43.7	37.8
		MCWB	33.6	33.0	37.9	46.0	53.7	58.5	64.1	63.0	57.6	47.0	37.2	32.9
	10%	DB	34.3	35.1	42.5	56.2	68.2	72.2	76.8	75.2	67.8	53.8	40.2	34.1
		MCWB	30.6	30.8	35.5	43.4	51.6	57.3	62.6	61.6	55.2	44.4	35.1	30.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.5	38.9	44.7	53.0	59.7	65.8	70.1	68.9	63.9	55.4	44.7	39.0
		MCDB	43.6	46.0	54.6	70.8	76.0	78.9	81.9	80.4	78.0	68.5	53.2	45.8
	2%	WB	36.0	35.6	40.8	49.6	57.1	62.3	67.8	66.2	60.5	51.0	40.8	35.7
		MCDB	40.7	40.7	49.1	64.7	72.2	74.0	79.8	78.6	74.0	61.4	46.9	40.6
	5%	WB	33.8	33.2	38.3	46.8	54.9	60.5	65.7	64.4	58.3	47.9	38.0	33.1
		MCDB	37.9	37.8	45.3	59.1	69.7	71.7	77.2	75.7	70.9	56.1	42.8	37.6
	10%	WB	30.6	30.8	36.0	44.2	52.9	58.8	63.8	62.5	56.0	45.4	35.5	30.2
		MCDB	34.1	34.9	41.7	54.8	66.2	69.4	74.2	73.1	66.1	52.4	39.6	34.2
Mean Daily Temperature Range	5% DB	MDBR	14.5	15.9	15.7	18.5	19.9	17.8	18.2	18.9	19.2	16.6	13.2	13.8
		MCDBR	17.1	19.8	20.9	26.7	26.1	23.2	23.2	24.4	27.7	24.4	17.5	17.0
		MCWBR	13.1	14.9	13.6	14.2	11.6	9.9	10.1	10.8	13.5	13.7	11.7	13.0
	5% WB	MCDBR	17.0	19.2	19.7	25.1	23.5	19.8	20.9	22.0	25.9	23.0	16.7	17.2
		MCWBR	13.4	14.9	13.3	14.0	11.0	9.6	10.6	11.0	12.9	13.6	11.7	13.7
Clear-Sky Solar Irradiance	taub	0.257	0.267	0.312	0.327	0.343	0.353	0.356	0.362	0.308	0.282	0.254	0.240	
	taud	2.307	2.356	2.308	2.389	2.392	2.380	2.418	2.351	2.499	2.547	2.456	2.315	
	Ebn at Noon	232	270	277	287	287	284	281	272	278	264	237	217	
	Edh at Noon	18	24	31	33	35	36	34	34	26	20	16	15	
All-Sky Solar Radiation	RadAvg	308	639	1129	1482	1817	1911	1955	1610	1111	639	306	227	
	RadStd	35	49	86	141	133	124	95	111	115	75	35	21	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+113	N/A
(47) Regional (68 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page